

MONTE CARLO SIMULATION OF THE HESTON MODEL

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$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^{(1)},$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma \sqrt{v_t} dW_t^{(2)}.$$

Exact identity (no approximation)

$$\log S_{t+\Delta t} = \log S_t - \frac{1}{2} \int_t^{t+\Delta t} v_s ds + \int_t^{t+\Delta t} \sqrt{v_s} dW_s^{(1)}.$$

Sources of numerical error

- Variance transition $v_{t+\Delta t}$
- Integrated variance $\int_t^{t+\Delta t} v_s ds$
- Monte Carlo sampling
(finite number of paths)

$$\text{Corr}(dW_t^{(1)}, dW_t^{(2)}) = \rho,$$

$$\text{Var}(dW_t^{(1)}) = \text{Var}(dW_t^{(2)}) = dt.$$

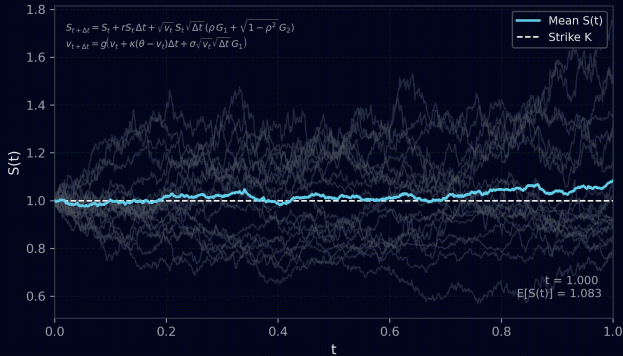
Scheme comparison

Method	$v_{t+\Delta t}$	$\int_t^{t+\Delta t} v_s ds$
Euler	Discretized	Discretized
Almost-Exact	Exact	Discretized
		Approximated

Key idea

- Almost-Exact removes bias from variance discretization
- Remaining error comes from time integration only
- Extra computational cost depends on parameters

Heston Model: Euler Monte Carlo Simulation



Heston Model: Exact variance + Almost-Exact terminal price

$$N \sim \mathcal{P}(\lambda), \quad v_{t+\Delta t} = \frac{\sigma^2(1 - e^{-\kappa\Delta t})}{2\kappa} \mathcal{G}(d + N), \quad \mathcal{G} = \text{Gamma},$$

$$S_1 = \exp(m + \Sigma G), \quad G \sim \mathcal{N}(0, 1).$$

- **Embarrassingly parallel MC:** 1 thread: 1 path (TOTAL_PATHS=262,144).
- **Per-thread RNG state:** one `curandState` per thread, initialized once.
- **Per-thread time loop:** propagate (S_t, v_t) for M steps, write payoff to **global memory**.
- **Reduction:** block-level sum in **shared memory** $\rightarrow$ 1 partial sum/block, final sum on host.
- **Step 2 change:** variance transition kernel via Poisson + Gamma sampling.

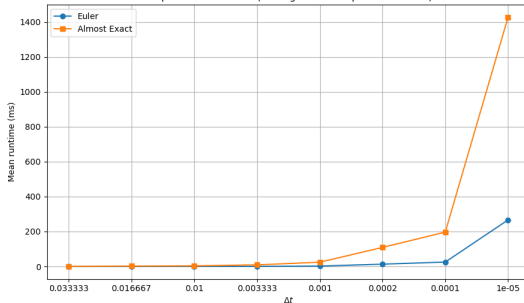
Representative run

Method	Price	t [ms]
Step 1 (Euler)	0.123081	2.385
Step 2 (Exact v_t)	0.123320	34.047

$$(S_0, v_0, r, \kappa, \theta, \sigma, \rho, T, K, M) = \\ (1, 0.1, 0, 0.5, 0.1, 0.3, 0, 1, 1, 1000) \\ \Delta t = 1/M.$$

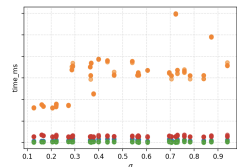
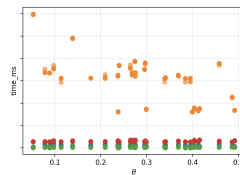
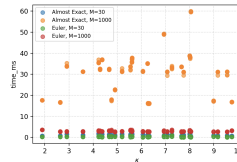
Step 3: Performance vs Δt and Parameters

Step 3: Runtime vs Δt (averaged over all parameter sets)

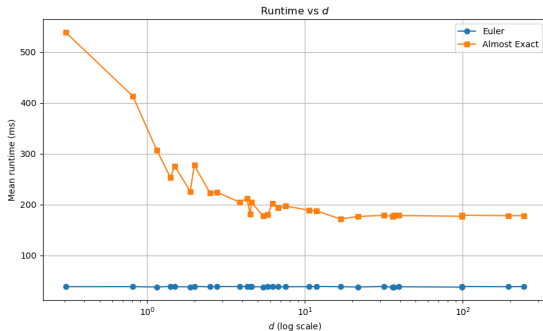


Average runtime per run [ms].

	$M = 30$	10^2	10^3	10^4	10^5
AE	0.998	3.565	24.48	196.7	1425.7
Euler	0.092	0.254	2.338	25.32	266.3



Step 3: Impact of Feller Condition on Elapsed Time



$$20 > \frac{\sigma^2}{\kappa \theta}$$

$$d = \frac{2\kappa\theta}{\sigma^2}$$

$$v_{t+\Delta t} = c\mathcal{G}(d + N)$$